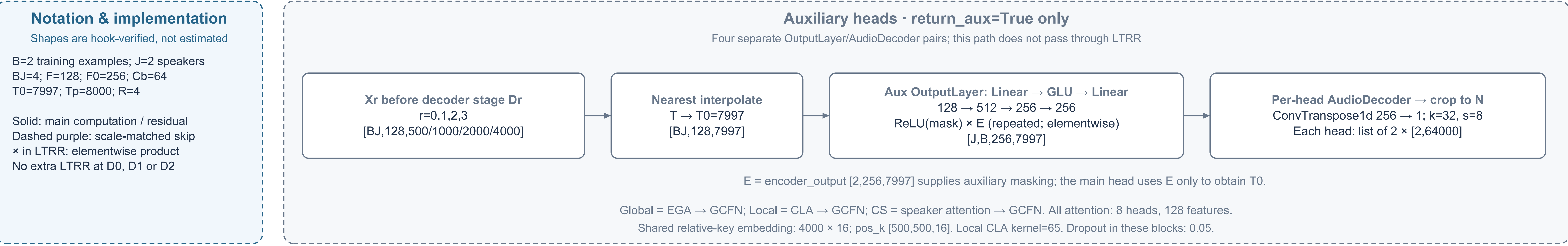
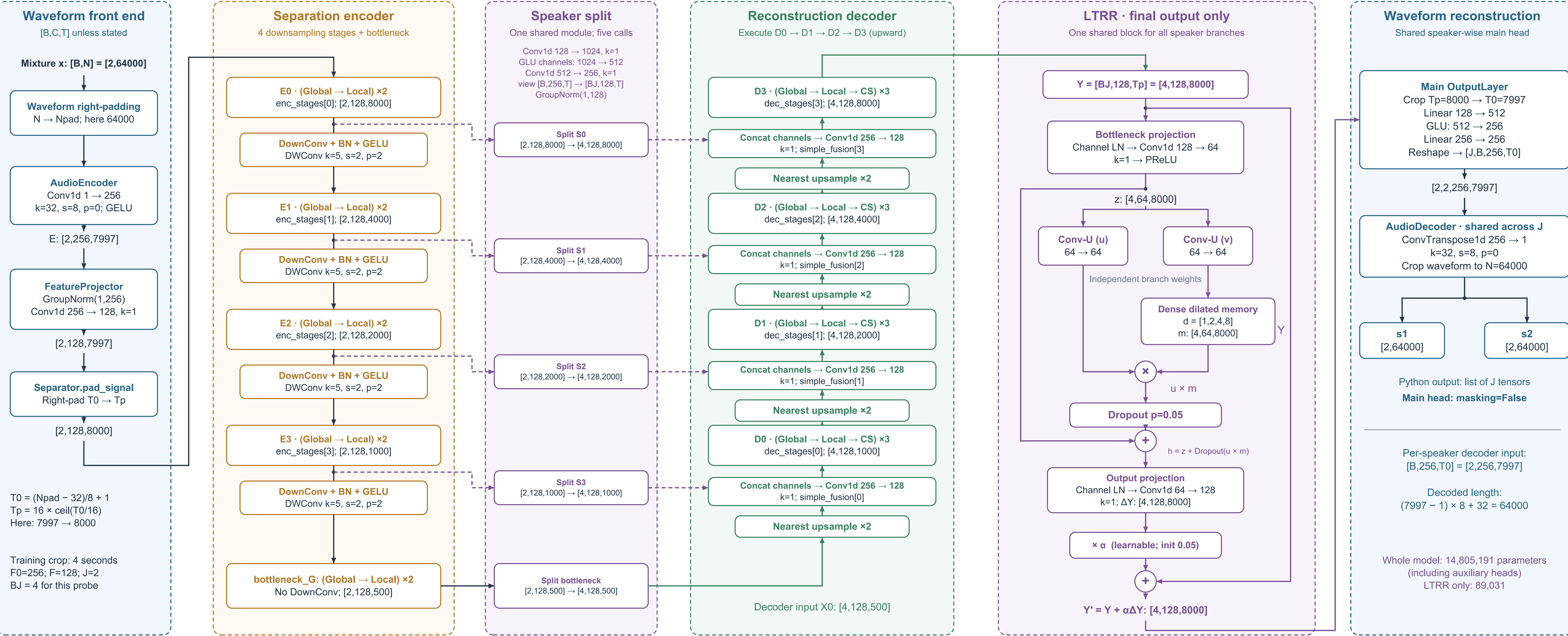


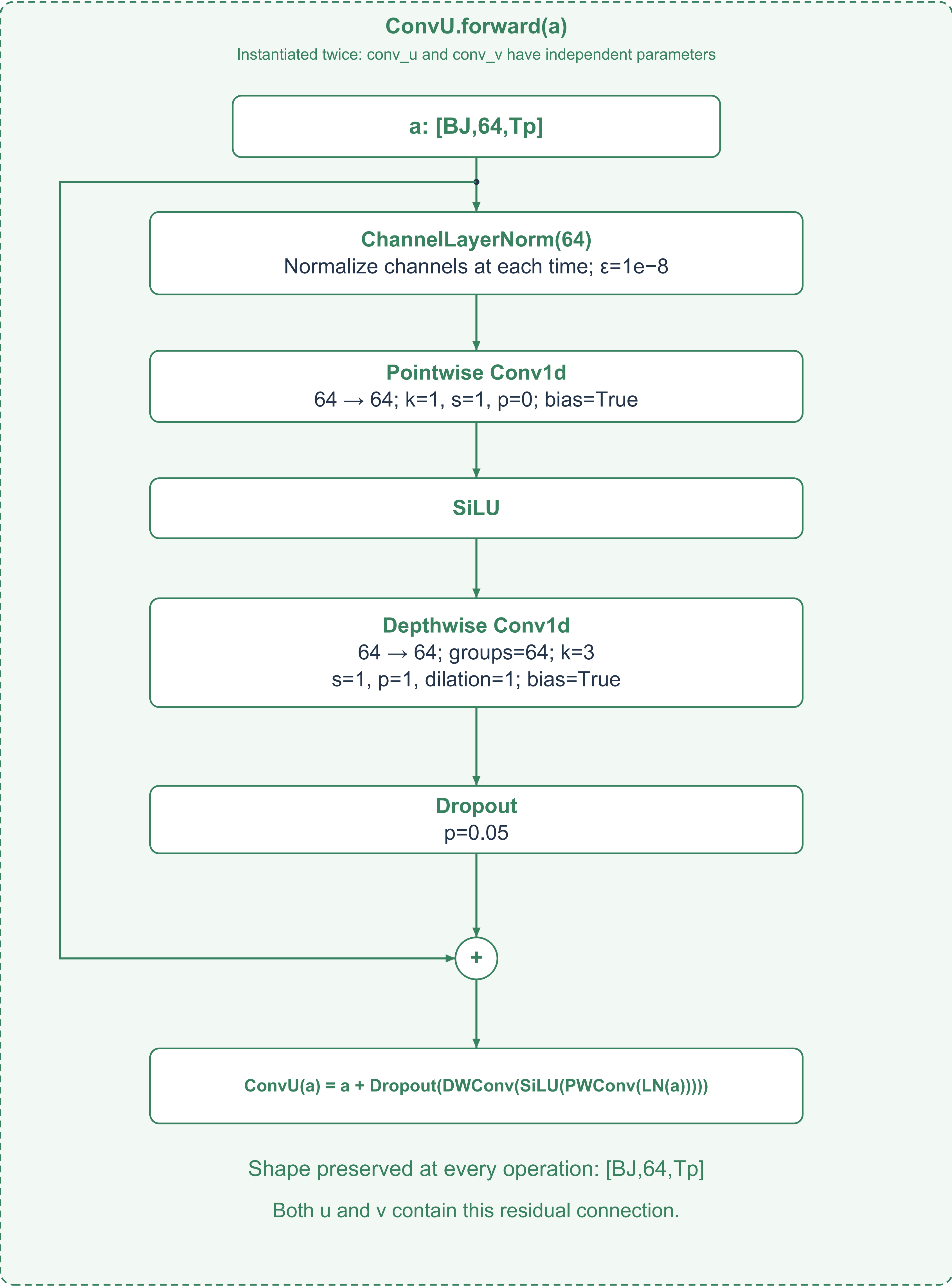
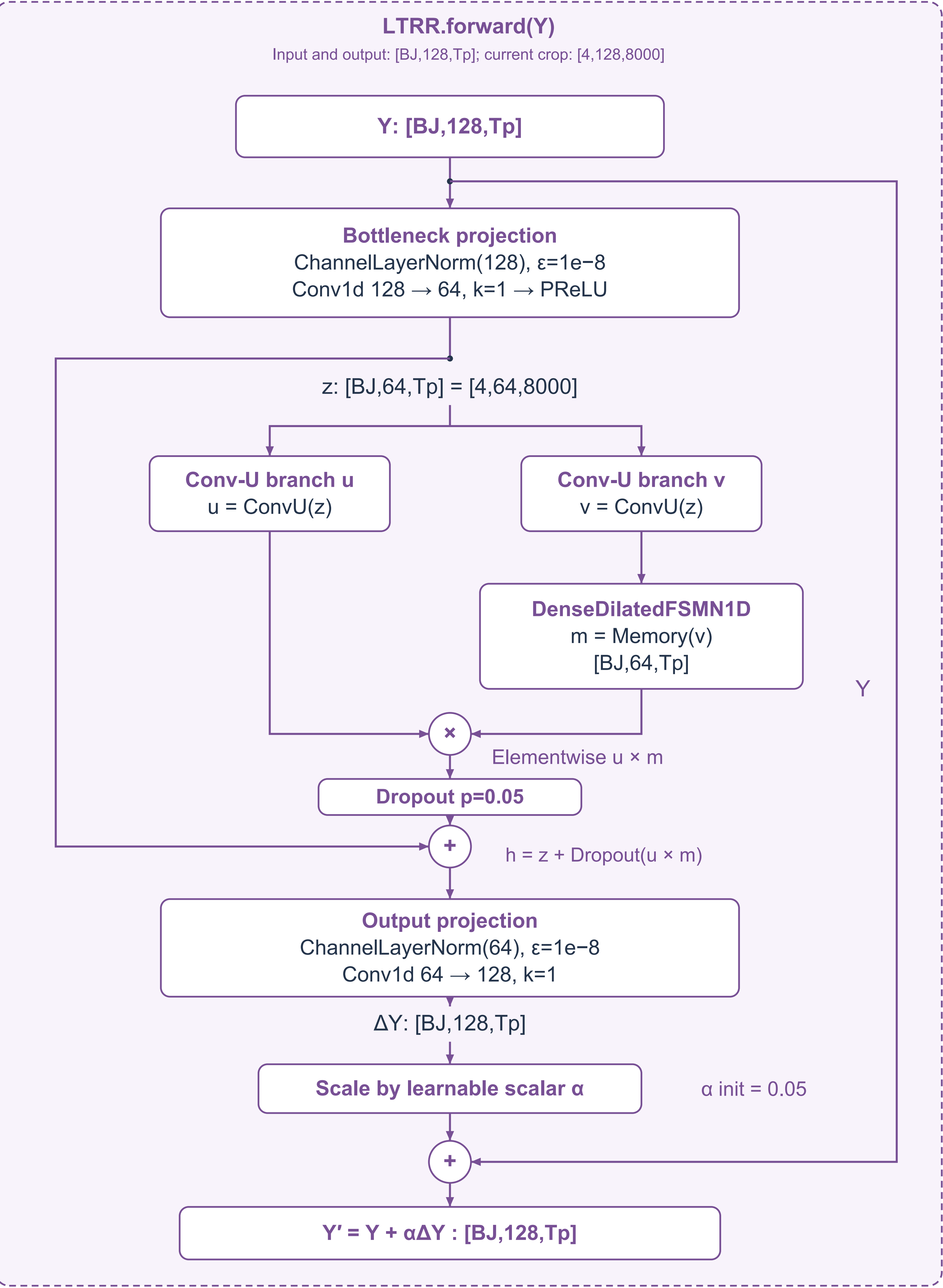
SepReformer + LTRR | detailed architecture

Current shared 16 kHz configuration · shape probe B=2, J=2, N=64,000 · stage indices follow Python ModuleList order



LTRR | residual refinement and the two Conv-U branches

Lightweight Temporal Residual Refinement · original active NoGate operations · only after the final reconstruction stage



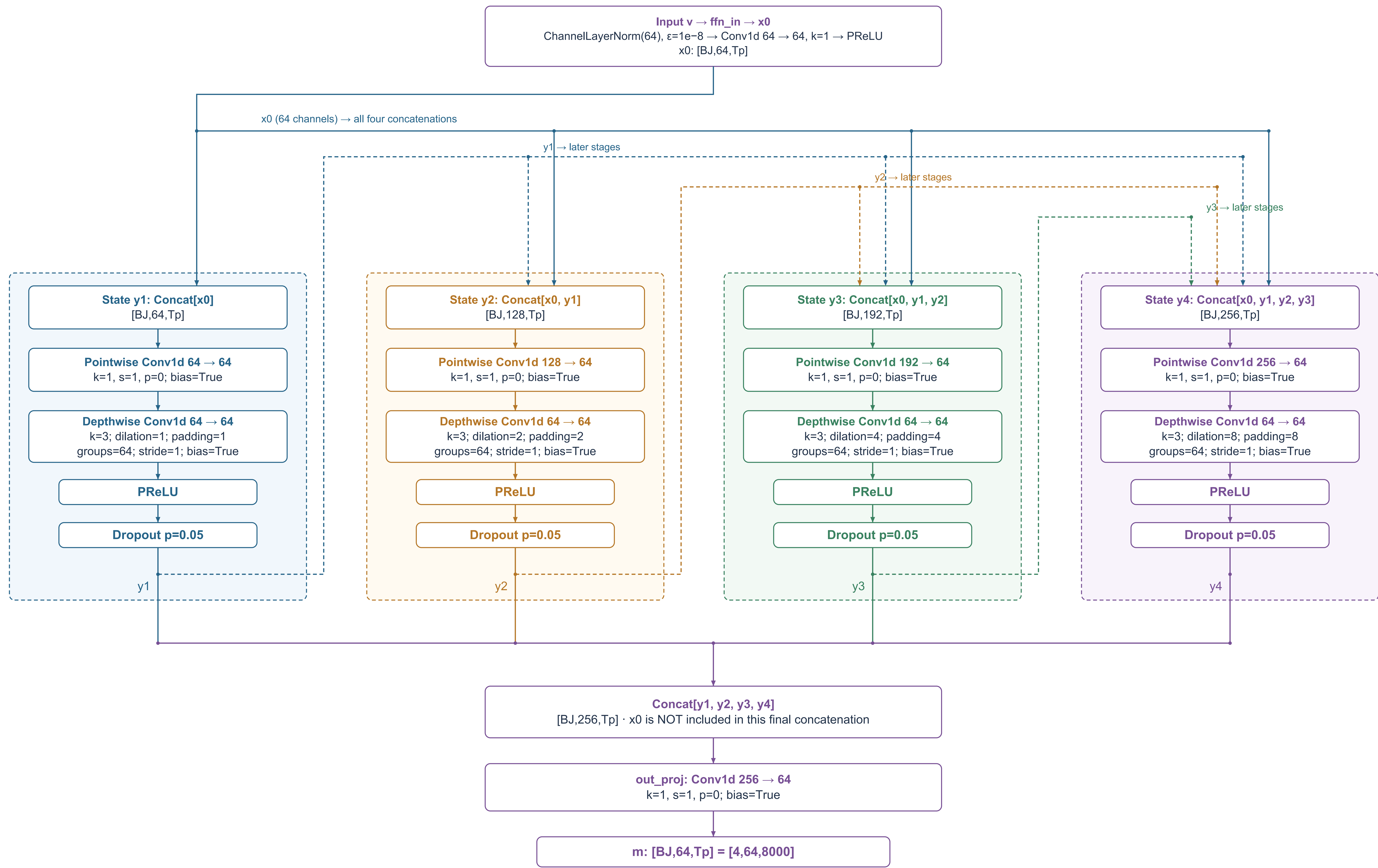
The outer bypass is Y; the inner bypass is z; each Conv-U has its own input bypass. Memory itself has no extra residual addition.

PReLU uses one learnable parameter (PyTorch default). LTRR convolutions have bias=True. No speaker-guided gate is instantiated.

Verified parameters: LTRR = 89,031; whole model including auxiliary heads = 14,805,191.

LTRR | DenseDilatedFSMN1D, unrolled exactly as implemented

Four sequential memory states · concatenation is along channels · every state has shape [BJ,64,Tp] = [4,64,8000]



Solid top bus: x_0 . Dashed feedback buses: earlier states to later stages. Dots denote junctions; crossings without dots do not connect.

All arrows follow the acyclic forward computation. The returned memory m is multiplied by u in TemporalInteraction; it is not added to v here.